Extensions of a nested ensemble Kalman filter

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Abstract

State-space models with a static parameter and a latent process make the observed-data likelihood an integral over a path. Nested particle filters estimate the successive parameter posteriors, but a particle likelihood becomes expensive as the observation dimension grows. An ensemble Kalman filter replaces that likelihood by a Gaussian forecast approximation and can be nested in the same outer recursion. We study how the resample-move step can be screened, switched and, if a different target is acceptable, weighted by a nearest-neighbour surrogate. On a linear Gaussian pilot, an exact tree returns the same neighbours as a list and evaluates fewer distances. A rule that uses plain Metropolis–Hastings while the series is short, and delayed acceptance afterwards, has a lower cost per accepted move than either kernel alone. Using the surrogate as the weight cuts the number of transition draws and increases the error in the posterior means. A Rao-Blackwellised correction leaves the weight unchanged when the observation model is already the linear Gaussian model inside the filter. A finite-difference Langevin proposal does not improve the effective sample size per draw. The comparison is a small Ornstein–Uhlenbeck pilot with a known likelihood.

Keywords: delayed acceptance, Ornstein–Uhlenbeck process, Rao-Blackwellisation, sequential Monte Carlo, surrogate likelihood

1 Introduction

State-space models are used when a latent Markov process is observed with noise. The parameter θ is static. The likelihood of a series is an integral over the latent path, and that integral is rarely available in closed form. Sequential Monte Carlo squared (SMC2) explores the sequence of parameter posteriors by weighting parameter particles with an unbiased particle-filter estimate of each likelihood increment, and by moving the particles with a particle Metropolis–Hastings kernel when the effective sample size falls (Chopin et al., 2012). The variance of that estimate grows with the length of the series and, more sharply, with the observation dimension.

The ensemble Kalman filter replaces the forecast distribution by a Gaussian distribution whose moments are the sample moments of an ensemble, and it updates the ensemble with a Kalman shift. The corresponding likelihood contribution is the Gaussian marginal of that forecast. It is not unbiased for the true likelihood. Golightly et al. (2026) place this contribution inside the SMC2 recursion and move the parameter particles with one step of ensemble Markov chain Monte Carlo. The resulting nested ensemble Kalman filter targets the parameter posterior induced by the ensemble Kalman likelihood, which we denote $\hat{\pi}_{\mathrm{enkf}}$, rather than the exact posterior.

The resample-move step is the expensive part of the recursion. At time t it reruns the ensemble filter from the start of the series for every proposal. Golightly et al. (2026) screen proposals with a k nearest-neighbour surrogate of the ensemble Kalman log-likelihood, and they note several further possibilities: a faster neighbour search, a choice among kernels, a gradient proposal, and the use of the surrogate in the weight as well as in the screen. They also give a Rao-Blackwellised weight for a nonlinear observation model. The discussion illustrates a regime

of small parameter dimension and larger observation dimension. Those remarks are the starting point of this paper. We do not treat them as theorems.

Our contribution is to state the target of each modification and to check the claims that can be checked on a linear Gaussian pilot. Firstly, the delayed-acceptance kernel is unchanged when a tree and a list return the same neighbours. Secondly, a choice among kernels that is made before the proposal, and that uses only the time index, preserves $\hat{\pi}_{\mathrm{enkf}}$ when each kernel does. Thirdly, a nearest-neighbour incremental weight targets the posterior under that surrogate, not $\hat{\pi}_{\mathrm{enkf}}$. Fourthly, when the true observation density equals the linear Gaussian density used by the filter, the Rao-Blackwellised weight equals the ensemble Kalman weight. A finite-difference approximation of the gradient of the ensemble Kalman log-likelihood can sit inside a corrected proposal; on the pilot it is not competitive per transition draw, and we do not recommend it.

The numerical section uses a scalar Ornstein–Uhlenbeck diffusion with a known Kalman likelihood. The pilot uses 60 parameter particles, 16 observations and an ensemble of size 12. It is not the experiment reported by Golightly et al. (2026).

2 Related work

SMC2 propagates a cloud of parameter particles. Each particle carries a particle filter. The incremental weight is an unbiased estimate of the likelihood contribution, and the θ-marginal of the extended target is the exact parameter posterior for any fixed number of state particles (Chopin et al., 2012, Proposition 1). The rejuvenation kernel is particle Metropolis–Hastings on that extended target. Replacing the inner particle filter by an ensemble Kalman filter removes the unbiasedness step. The marginal is then $\hat{\pi}_{\mathrm{enkf}}$.

Ensemble Markov chain Monte Carlo uses the same ensemble Kalman likelihood inside one Metropolis–Hastings chain on a fixed data set (Drovandi et al., 2022). One rejuvenation sweep of the nested filter is one iteration of that scheme. The chain is not sequential. Even when the forecast is exactly Gaussian, the plug-in Gaussian density is a biased estimator of the idealised Gaussian predictive, and an unbiased estimator of that Gaussian density does not restore the exact posterior of the state-space model.

A different sequential construction gives the parameter an artificial transition and resamples it with a shrinkage kernel, so that the filter does not collapse onto a single parameter value (Liu and West, 2001). The particle ensemble Kalman filter of Katzfuss et al. (2018) weights parameter particles by an integrated ensemble Kalman likelihood and, for a constant parameter, points back to that kernel because a point-mass proposal would otherwise leave fewer and fewer distinct particles. The nested filter keeps θ static and restores diversity by a Metropolis–Hastings move.

Delayed acceptance screens a proposal with a cheap approximation and corrects it with the intended target (Christen and Fox, 2005). If the parent kernel is irreducible for that target, the proposal is reversible, and the approximation is positive wherever the proposal is positive, the target remains invariant. Sherlock et al. (2016) store the expensive evaluations in a KD-tree and use an inverse-distance neighbour average as the screen, in batch Markov chain Monte Carlo. Bon et al. (2021) put a surrogate inside the mutation of an SMC sampler for a static model. In both cases the second stage evaluates the expensive target. Using the surrogate as the target is a different algorithm (Sherlock et al., 2016, Section 5.1).

Katzfuss et al. (2018) give the variance of the log ensemble Kalman likelihood as of order n/N in one independent linear example, when the forecast members are drawn independently from the true forecast (their Proposition 2). That is the scaling comparison cited by Golightly et al. (2026). It is not a theorem about the nested filter. Temfack and Wyse (2026) also replace the inner particle filter of SMC2 by an ensemble Kalman filter, with a state-dependent observation variance and an unbiased Gaussian density estimator, and they are explicit that the Gaussian assumption remains. Their mutation is particle marginal Metropolis–Hastings. They do not use

delayed acceptance or the Rao-Blackwellised weight studied here.

Adaptive SMC can choose among kernels that each leave the current target invariant, treating the tuning as an auxiliary variable and scoring kernels by expected squared jumping distance (Fearnhead and Taylor, 2010). A later SMC2 construction switches between particle marginal Metropolis–Hastings and particle Gibbs (Botha et al., 2023). Particle Gibbs needs a conditional particle filter. An ensemble Kalman filter does not supply one, and the unbiasedness used in those arguments is not available for $\hat{\pi}_{\mathrm{enkf}}$. We use only the idea of a choice made before the proposal.

Chen et al. (2021) differentiate the ensemble Kalman log-likelihood by automatic differentiation and ascend on it. In the linear Gaussian case they prove an L^p rate of $N^{-1/2}$ for that gradient. The ascent has no accept-reject step. A particle-filter gradient inside an uncorrected Langevin proposal (Rosato et al., 2024) is a derivative of a different likelihood. We do not use either proposal as a method. Section 4 reports a corrected finite-difference proposal and the reason it is not retained.

3 Methods

The observation model in the linear Gaussian case is

$$f_t^{(\theta)}(y_t \mid x_t) = \mathcal{N}(y_t; Hx_t, R). \tag{1}$$

An ensemble of size N is propagated by the state transition. The forecast mean $\hat{\mu}_{t|t-1}$ and covariance $\hat{\Sigma}_{t|t-1}$ are the sample moments, with divisor $N-1$. The ensemble Kalman likelihood contribution is

$$\hat{p}^{(\theta)}_{\mathrm{enkf},t}(y_t \mid y_{0:t-1}) = \mathcal{N}\big(y_t; H\hat{\mu}_{t|t-1},\, H\hat{\Sigma}_{t|t-1}H^\top + R\big). \tag{2}$$

This is exact for the Gaussian forecast approximation. It is not unbiased for the true predictive density. The product of these contributions, multiplied by the prior, defines $\hat{\pi}_{\mathrm{enkf},t}$ up to the ensemble randomness that the extended target carries (Golightly et al., 2026, Section 4.1).

3.1 Nested filter

The outer recursion follows Algorithm 6 of Golightly et al. (2026). At time t, each parameter particle is updated by one ensemble Kalman step and its weight is multiplied by (2). If the effective sample size is at least γM, the cloud is left as it is. Otherwise the particles are resampled with replacement, the weights are reset to $1/M$, and each particle is offered one Metropolis–Hastings proposal. The proposal is a random walk whose covariance is ζ times the sample covariance of the other resampled particles. The acceptance ratio uses the prior and the ensemble Kalman likelihood of the series so far. The likelihood at a proposed value is computed by rerunning the filter from the known initial condition. In the pilot, $\zeta^2 = 2.56^2/d_\theta$ and the ensemble size is held fixed.

The printed Algorithm 6 places the effective-sample-size test inside the particle loop. The surrounding text applies one resample-move step when the effective sample size of the updated cloud falls below γM. We follow that reading.

3.2 Delayed acceptance and the neighbour search

The screen replaces both ensemble Kalman likelihoods in the acceptance ratio by an inverse-distance average of the k nearest stored log-likelihoods (Golightly et al., 2026, Section 4.2). If the screen accepts, a second ratio corrects back to the ensemble Kalman likelihood. The product of the two ratios equals the plain Metropolis–Hastings ratio, so the kernel leaves the extended ensemble Kalman target invariant under the conditions of Christen and Fox (2005).

The distance in the paper is described as some metric. The pilot uses Euclidean distance on $\log \theta$, after standardising each coordinate by the standard deviation of the stored cloud.

A linear scan and an exact KD-tree that return the same neighbours produce the same surrogate and the same kernel. A different deterministic neighbour set changes the kernel and, if the surrogate is positive wherever the proposal is positive, still leaves the ensemble Kalman target invariant, because the second stage evaluates that target. Approximate neighbours are not used. At the pilot size the question is only whether the tree evaluates a distance to fewer stored points. The tree is rebuilt on every query, so the count is not a claim about wall-clock time.

3.3 A switch that depends only on time

While t is small, the ensemble Kalman likelihood is a short filter and a screen saves little. The pilot rule uses plain Metropolis–Hastings when $t \leq 8$ and delayed acceptance with $k = 3$ when $t > 8$, on a series of length 16. The choice is a function of t only and is made before the proposal. Each kernel preserves $\hat{\pi}_{\mathrm{enkf},t}$. A mixture of kernels with the same invariant distribution preserves that distribution. The argument does not use the unbiasedness hypothesis in Chopin et al. (2012) or the particle-Gibbs kernel in Botha et al. (2023).

The criterion, fixed before any output was seen, is the number of transition draws used inside rejuvenation divided by the number of accepted proposals. That comparison is in the same spirit as the relative cost of the cheap approximation and the stage-two acceptance rate in the delayed-acceptance efficiency analysis of Sherlock et al. (2021). We do not use their diffusion limit.

3.4 Surrogate incremental weights

If the incremental weight is the nearest-neighbour prediction of (2), the weights are aimed at

$$\pi_{\mathrm{knn},t}(\theta) \propto \pi(\theta)\, \exp\bigl(\ell_{\mathrm{knn},t}(\theta)\bigr), \tag{3}$$

conditional on the stored cloud that defines $\ell_{\mathrm{knn},t}$. This is not $\hat{\pi}_{\mathrm{enkf},t}$ and not the exact posterior. The two errors are separate. Equation (2) already differs from the true likelihood. The surrogate then differs from (2).

In the pilot the first k particles are a reference set and are always updated with the ensemble Kalman filter. Every other particle takes the inverse-distance prediction of the reference increments and does not run a forecast for that weight. After resampling, the reference slots are re-evaluated so that the screen is built from ensemble Kalman likelihoods. A proposal that passes the screen is still evaluated with the ensemble filter. There is no bound on the distance between $\pi_{\mathrm{knn},t}$ and $\hat{\pi}_{\mathrm{enkf},t}$. The diagnostic, also fixed in advance, recomputes (2) for eight non-reference particles at the terminal time. A mean, over replicates, of the median absolute log difference above 1 was the threshold for calling the surrogate a poor approximation of the ensemble Kalman increment.

3.5 Rao-Blackwellised weight

For a general observation density f, the filter can still be run under a linear Gaussian approximation g. The Rao-Blackwellised increment averages, over the analysis ensemble,

$$\hat{g}^{(\theta)}_{\mathrm{enkf},t}(y_t)\, \frac{f^{(\theta)}_t(y_t \mid x^j_t)}{g^{(\theta)}_t(y_t \mid x^j_t)}. \tag{4}$$

The parameter target induced by these increments is denoted $\tilde{\pi}_{\mathrm{enkf},t}$. If $f = g$, the ratio equals 1 for every member, and (4) equals (2). The correction does not change the target and the extra

Table 1: Kernel comparison. Means over four replicates. Cost per acceptance is the number of rejuvenation transition draws divided by the number of accepted proposals. Source: `summary-s2.csv`.

Kernel	Abs. error	Draws	Cost per acceptance	ESS
Metropolis–Hastings	0.172	33855	310.4	39.6
Delayed acceptance	0.126	23970	221.6	43.3
Switch at $t = 8$	0.132	23457	214.7	36.7

density evaluations do not change the estimate. The identity is linear and Gaussian. It is not a general rule for a nonlinear observation model.

Worth its cost, fixed before the runs, means a smaller absolute error of the posterior means of $\log \theta_i$, per transition draw, than the plain filter that uses the same g. When $f = g$ the identity says the correction cannot improve the error.

4 Simulation studies

The latent process is the scalar Ornstein–Uhlenbeck diffusion in Section 5.1 of Golightly et al. (2026),

$$\mathrm{d}X_t = \theta_1(\theta_2 - X_t)\,\mathrm{d}t + \theta_3\,\mathrm{d}W_t,$$

with the Gaussian transition printed there. The truth is $\theta = (1, 2, 1)$, the initial condition is $x_0 = 10$, and observations at times $1, \ldots, 16$ use (1) with $H = 1$ and $R = 0.1$. The prior is independent, $\theta_1 \sim \text{Gamma}(2, 2)$, $\theta_2 \sim \text{Gamma}(5, 3)$ and $\theta_3 \sim \text{Gamma}(2, 5)$, in the shape-rate parameterisation. The PDF writes $\text{Gamma}(a, b)$ without naming the parameterisation. We use shape and rate.

The nested filter uses $M = 60$, $N = 12$, $\gamma = 0.4$ and $k = 3$. The ensemble size does not adapt. Four replicate series are used. The gold standard for each series is a random-walk Metropolis–Hastings chain on $\log \theta$, targeting the exact Kalman likelihood, with 3000 iterations after 1000 burn-in and proposal standard deviation 0.15. That chain is much shorter than the 10^6 iterations used by Golightly et al. (2026), and M, T and N are smaller than in their Section 5.1. Posterior means from the pilot must be read with that Monte Carlo error in mind.

The reported error is the absolute difference, averaged over the three coordinates, between the weighted mean of $\log \theta_i$ and the gold-standard mean. Cost is the number of transition draws. Tables 1–3 give means over the four replicates. The files in the accompanying results directory contain the replicates.

On the pre-registered criterion the switch is the best of the three kernels. Delayed acceptance is close to it and better than the plain random walk. The neighbour sets from the list and the tree agreed on all 491 checks per replicate on average. The list evaluated a distance to 29460 stored points on average and the tree to 13422.5. The agreement is the result that keeps the kernel unchanged. The distance count favours the tree at this size and does not include the cost of building the tree.

Figure 1 places the same runs on one scale, together with the surrogate-weight filter and the finite-difference proposal. The finite-difference proposal had mean absolute error 0.074 and mean transition draws 268602, against 0.172 and 33855 for the random walk. Effective sample size per transition draw was 0.000148 against 0.00123. The proposal does not improve that ratio. We do not retain it. The acceptance probability included the Gaussian proposal density, so the negative result is about cost, not about leaving $\hat{\pi}_{\text{enkf}}$.

The surrogate-weight filter used 5217 transition draws on average. The mean of the median absolute log differences between the surrogate increment and a fresh ensemble Kalman increment was 0.277, below the threshold of 1. The mean absolute error was 0.293, larger than the error

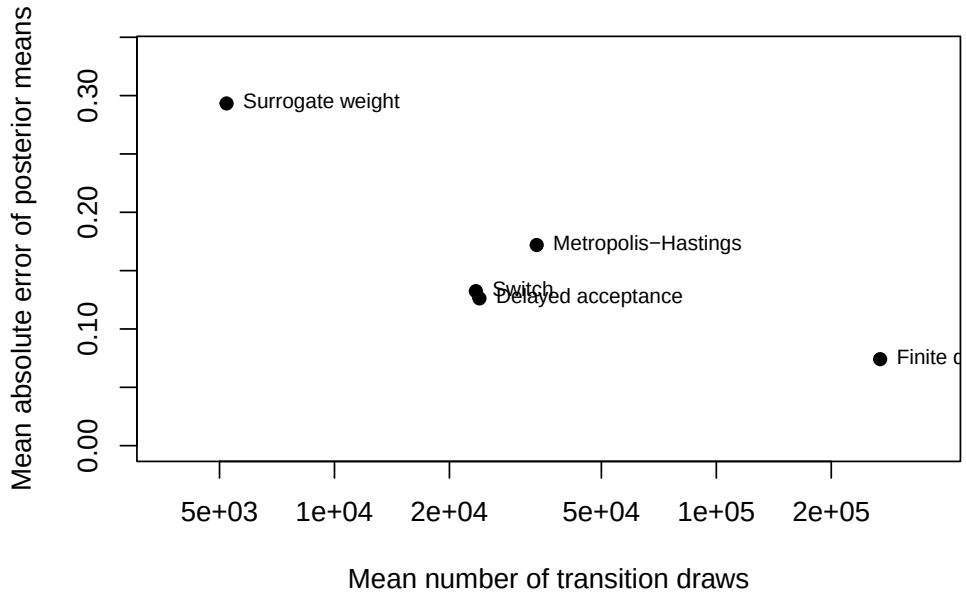


Figure 1: Mean absolute error of the posterior means of $\log \theta_i$ against the mean number of transition draws. The finite-difference proposal is shown as a comparison that is not retained. Source: `summary-s2.csv`, `summary-s3.csv` and `summary-s4.csv`.

Table 2: Sample variance of the log ensemble Kalman likelihood at the true parameter, from 20 independent ensembles. Source: `s5-variance.csv`.

d_y	N	Variance	Mean draws
1	8	14.060	128
1	16	2.879	256
5	8	24.935	640
5	16	10.661	1280

of the plain filter and of delayed acceptance. The cost falls because most particles do not run a forecast. The target is (3). The larger error is the visible part of the second approximation.

Table 2 is the scaling check that this pilot can support. The five-dimensional series is five independent copies of the same diffusion, with independent observation noise. Doubling N reduces the variance in both rows. The variance does not scale by the ratio of the observation dimensions. Katzfuss et al. (2018, Proposition 2) assumes independent draws from the true forecast in a different example. The ensemble used here is interacting, and the log-likelihood is a sum over the series. We do not draw a boundary at parameter dimension 10 or observation dimension 10.

When $R_g = R$, the maximum absolute difference between the log ensemble Kalman increment and the log Rao-Blackwellised increment was 0 on each replicate. The correction does not change the weight. When $R_g = 0.5$ and the true observation variance is 0.1, both densities are Gaussian and $f \neq g$. The Rao-Blackwellised filter then has the smaller error per draw. That comparison is not a nonlinear observation model, and SMC2 was not run.

Table 3: Rao-Blackwellised weights. The identity row uses $R_g = R = 0.1$. The other two rows use $R_g = 0.5$ whilst the data and the gold standard use $R = 0.1$. Error per draw is the ratio of the two means. Source: `summary-s6.csv` and `s6-identity.csv`.

Filter	Abs. error	Draws	Error per draw
Plain, $R_g = 0.5$	0.369	22776	1.62×10^{-5}
Rao-Blackwellised, $R_g = 0.5$	0.169	36069	4.68×10^{-6}
Rao-Blackwellised, $R_g = 0.1$	0.121	31248	3.87×10^{-6}

5 Discussion

The nested ensemble Kalman filter is a practical way to keep an SMC2 recursion when an unbiased particle likelihood is too expensive, provided one is willing to target $\hat{\pi}_{\text{enkf}}$ rather than the exact posterior. The pilot supports three concrete modifications and rejects one.

Delayed acceptance, with a neighbour surrogate that is corrected by the ensemble Kalman likelihood, reduced both the number of transition draws and the error relative to a plain random walk, on this linear Gaussian example. An exact tree did not change the neighbour sets. It evaluated fewer distances than a list at parameter dimension 3 and 60 particles. Whether that reduction survives the cost of maintaining the tree, and whether it survives a larger cloud, is open. Approximate neighbours were not given an invariance proof, and they were not used.

Choosing the plain kernel for the first half of the series and delayed acceptance afterwards improved the cost per accepted move slightly beyond delayed acceptance alone. The rule is crude. It is a function of time, which is enough for the mixture argument. A rule that looked at the proposed particle would need a different justification, and we did not use one.

Weighting by the surrogate changes the target. On the pilot the diagnostic stayed inside the threshold we had set, the number of transition draws fell by a large factor, and the posterior means moved further from the gold standard than the means under $\hat{\pi}_{\text{enkf}}$. We would use the surrogate weight only when that extra error is acceptable. We do not have a bound on it.

The Rao-Blackwellised correction is idle when the observation model is already the linear Gaussian model inside the filter. The identity is exact in the pilot, up to the reported difference of 0. When the filter is run under a deliberately inflated Gaussian variance, the correction reduced the error per draw. That is a narrow illustration. It does not establish the ordering, stated by Golightly et al. (2026), between the plain nested filter, the Rao-Blackwellised filter and SMC2 on a nonlinear observation model.

The finite-difference proposal is the modification we drop. A derivative of the ensemble Kalman log-likelihood exists in the linear Gaussian case (Chen et al., 2021), and a Metropolis–Hastings correction can use a noisy finite difference without changing $\hat{\pi}_{\text{enkf}}$. Forming the difference, and the reverse density, repeated the filter often enough that the effective sample size per transition draw fell. An uncorrected Langevin proposal was not used.

Several limits are specific to the pilot. The gold-standard chain is short. The parameter dimension is three and the observation dimension in the main comparison is one. The five-dimensional variance table is the only step towards the regime suggested by Golightly et al. (2026), and it does not confirm a threshold. Gamma priors were given a shape-rate reading that the source paper does not state. A real series was not analysed: the comparisons that the pilot can make already have a known likelihood, and the incidence series we considered are not high-dimensional observation vectors.

What remains open is a bound for the surrogate posterior, a neighbour search whose wall-clock cost is measured at the cloud sizes used in applications, and a nonlinear observation model on which the Rao-Blackwellised weight can be compared with SMC2. Those are the claims this pilot does not make.

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